

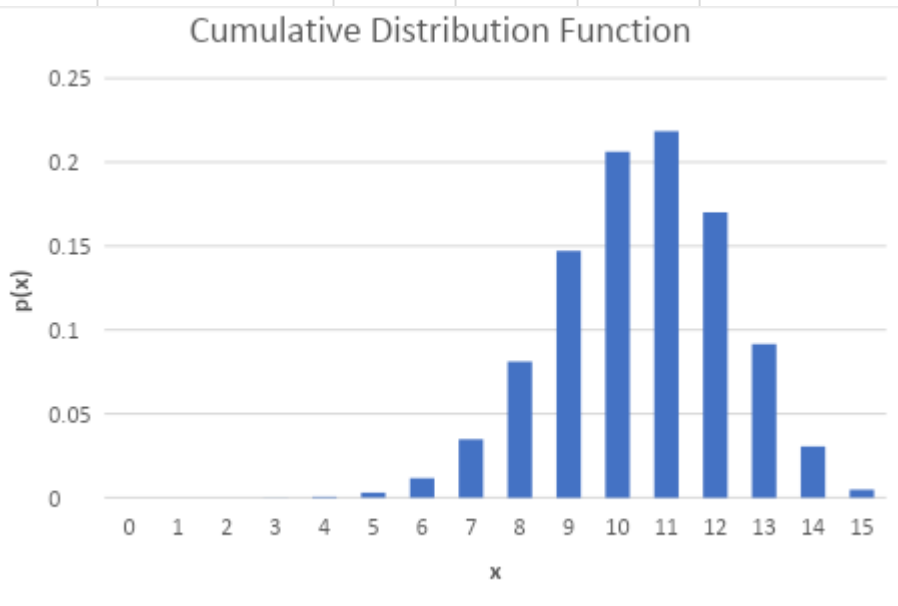
BINOMIAL DISTRIBUTION

No. of Trials, $n=15$

Mean= $n.p$

Variance=?

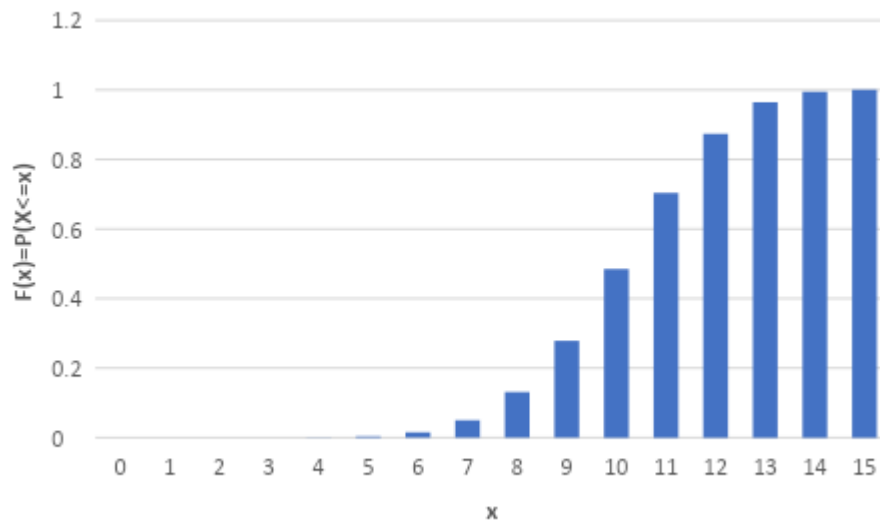
x	$p(x)$
0	0.00000001434890
1	0.00000050221174
2	0.00000820279183
3	0.00008293933967
4	0.0005805753777
5	0.002980286939
6	0.01159000476
7	0.03477001428
8	0.08113003333
9	0.1472359864
10	0.206130381
11	0.2186231313
12	0.1700402133
13	0.09156011483
14	0.03052003828
15	0.00474756151



Probability of Success, $p=0.7$

x	$F(x)=P(X \leq x)$
0	0.000000014348907
1	0.000000516560652
2	0.000008719352487
3	0.00009165869215
4	0.0006722340698
5	0.003652521008
6	0.01524252577
7	0.05001254005
8	0.1311425734
9	0.2783785598
10	0.4845089408
11	0.7031320721
12	0.8731722854
13	0.9647324002
14	0.9952524385
15	1

Probability Mass Function

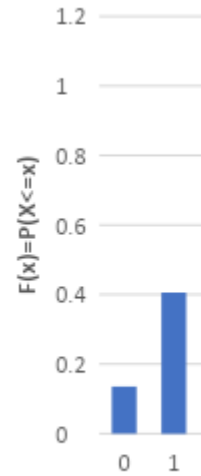
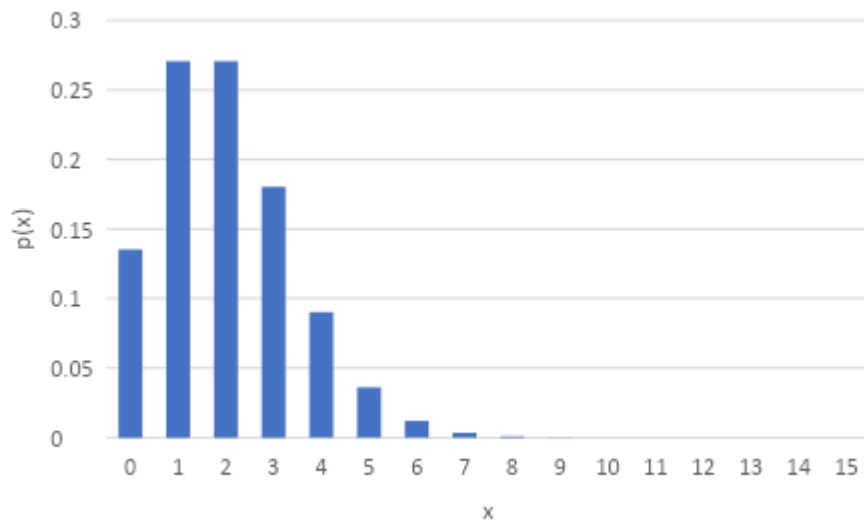


POISSON'S DISTRIBUTION

Mean(μ)=2

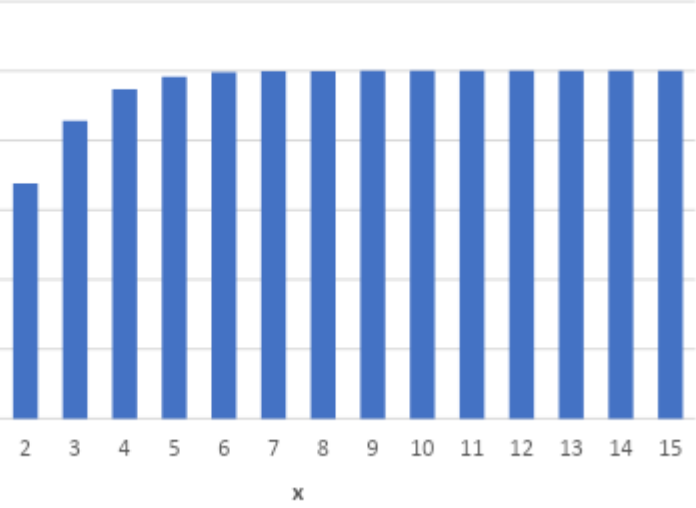
x	p(x)	x
0	0.135335283236613	0
1	0.270670566473226	1
2	0.270670566473226	2
3	0.180447044315484	3
4	0.090223522157742	4
5	0.036089408863097	5
6	0.012029802954366	6
7	0.003437087318133	7
8	0.000859271829533	8
9	0.000190791284341	9
10	0.000038158256868	10
11	0.000006908486196	11
12	0.000001151414366	12
13	0.000000191902394	13
14	0.000000031983732	14
15	0.000000005327111	15

Cumulative Distribution Function



F(x)=P(X<=x)			
0.1353352832			
0.4060058497			
0.6766764162			
0.8571234605			
0.9473469827			
0.9834363915			
0.9954661945			
0.998903281			
0.9997625527			
0.9999535019			
0.9999916918			
0.9999986354			
0.9999997927			
0.9999999707			
0.9999999961			
0.9999999995			

Probability Mass Function



<u>FITTING OF BINOMIAL DISTRIBUTION</u>				
Q. Find a Binomial Distribution to the following data.				
No. of Heads	1	2	3	
Frequency	4	13	28	
<u>x</u>	<u>OBSERVED FREQUENCY</u>	<u>x*f</u>	<u>p(x)</u>	
1	4	4	0.03717608169	
2	13	26	0.1291694422	
3	28	84	0.2493351165	
4	42	168	0.2887741835	
5	20	100	0.200670961	
6	6	36	0.07747082684	
7	2	14	0.01281784651	
<u>TOTAL</u>	115	432	0.9954144582	
	Mean(μ)	3.75652174		
	n	7		
	p	0.53664596		
	N	115		

4	5	6	7
42	20	6	2
Expected Frequency			
4.275249395			
14.85448585			
28.6735384			
33.2090311			
23.07716051			
8.909145087			
1.474052348			
114.4726627			

